

WTW 158 - CALCULUS - Study guide 2025

A. Organization

1.	Admittance	2
2.	Textbook	2
3.	Contact details	2
4.	Communication	2
5.	Lectures	2
6.	Tutorials	3
7.	Consultation	3
8.	Announcements	3
9.	Calculators	3
10.	Study time	3

B. Assessment and related matters

1.	Material for class tests, semester tests and the exam	3
2.	Absence from class tests, semester tests or the exam	4
3.	Semester test dates	5
4.	Students with special needs	5
5.	Marked tests and test queries	5
6.	Mark list	5
7.	Semester mark	5
8.	Entry requirement for the exam	5
9.	Final mark	5
10.	Supplementary examination	6
11.	Disciplinary cases	6

C. Course contents

1.	Use of the study guide	7
2.	General objectives	7
3.	Module outcomes	7
4.	Module structure	7

Theme 1:	Functions	8
Theme 2:	Limits, continuity and derivatives	15
Theme 3:	Differentiation	21
Theme 4:	Application of differentiation	25
Theme 5:	Integration	30
Theme 6:	Vector algebra	33

Appendix:	Technique mastering	36
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The only way to learn mathematics, is to do mathematics

Paul Halmos

A. Organization

1. Admittance

You may register for this module only if the official class timetable allows you to attend all the classes and a tutorial and if there are no test or examination timetable clashes. Any exceptions must be approved by the dean concerned.

2. Textbook

Calculus Early Transcendentals, 9th Edition, J Stewart, D Clegg and S Watson.
ISBN: 978-0-357-11351-6.

3. Contact details

Module coordinator: Dr AS Jooste

Course Administrator: Dr T Tegegn.

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Lecturer	Dr T Rakotonarivo	Maths 2-12	-	tsinjo.rakotonarivo@up.ac.za
Lecturer	Dr T Tegegn	Maths 2-23	-2485	tesfalem.tegegn@up.ac.za

4. Communication

Please always use proper English and full sentences when communicating with your lecturers or fellow students. When you send an email, remember the following:

- (a) Your email should be addressed to one person only. You may copy a second person, e.g., your lecturer or the module coordinator;
- (b) We expect you to use a respectful tone in formal English and the correct punctuation marks. Disrespectful emails will be ignored;
- (c) The course email address: WTW158@up.ac.za is for administrative issues only (including time table problems);
- (d) Mathematical questions should be directed to your lecturer;
- (e) Insert a clear and explanatory subject in the subject line, including the module code: e.g., 'WTW 158 Class attendance - P Buys';
- (f) Your email should be short and clear;
- (g) Please start your email formally, using the correct form of address, e.g., Dear Prof Kufakunesu;
- (h) At the end of your message, insert your full name, surname and student number.

5. Lectures

Number of lectures per week: 4. Class attendance is compulsory for all students for the full duration of the course. A student may be refused admission to the examination or promotion to a subsequent year of study if he/she fails to comply with the attendance requirements.

6. Tutorials

You will be assigned to a tutorial group within the first two weeks of the semester. You have to attend one tutorial class per week. Each tutorial class addresses material covered in the lectures of the previous week, **as indicated on the lecture schedule on ClickUP**. You should study the theory and attempt the tutorial problems, as given in the study guide, before the relevant tutorial. Tutorial time is important and should be seen as an opportunity to sort out difficulties/uncertainties you have with the scope covered the previous week, therefore it is important to **prepare beforehand**.

Short class tests will be written during tutorials. No extra time will be given for class tests.

Tutorial attendance is compulsory. A student may be refused admission to the examination or promotion to a subsequent year of study if he/she fails to comply with the attendance requirements.

7. Consultation

Each lecturer has consultation hours, that will be displayed on ClickUP. Students may consult lecturers only during these hours, or by appointment made by email. We recommend that you make use of this opportunity as soon as you experience problems.

8. Announcements

Announcements will be made on ClickUP; you should read these regularly.

9. Calculators

No calculators are allowed.

10. Study time

You need to spend 12 hours per week on this module. The scheduled contact time is about 6 to 7 hours per week, which means that another 5 to 6 hours per week of own study time should be devoted to this module.

B. Assessment and related matters

The examination and test instructions of the University of Pretoria must be followed meticulously. **Please note that when you enter the exam/test venue, you declare yourself fit to write the particular exam/test.** See General Rules and Regulations

1. Material for class tests, semester tests and the exam

You will be informed beforehand of the scope of each class test/semester test/exam.

In the class tests, semester tests and exam, there will be a mixture of questions designed to test:

- (a) your understanding of basic ideas, definitions and theory,
- (b) your ability to perform routine calculations, and

(c) the depth of your understanding of the module material by proving unseen results and solving unseen problems.

2. Absence from class tests, semester tests or the exam

- If you are absent from a **class test** for a valid reason, then that test will be excluded when your class test mark is calculated. However, a mark of 0 will be assigned for any further class tests that you miss, for whatever reason. The Calculus Readiness Test will be considered as your first class test and it is compulsory to write this test.
- If you know in advance that you will be unable to attend a **semester test** for a valid reason, you must notify the course administrator (WTW158@up.ac.za) timeously by email so that an alternative arrangement can be made. Failure to do so, will result in you being assigned a mark of 0 for the test in question.

If you are absent from a semester test you must notify the course administrator (WTW158@up.ac.za) by email within three (working) days of the test being written and offer a valid reason (see below) for your absence. Original documentation (e.g., sick notes) must be handed in at the office of the course administrator. Failure to follow the instructions, will result in you being assigned a mark of 0 for the test in question.

If you cannot write a test on a Saturday, due to religious reasons, you need to follow the guidelines in General Rules and Regulations A14

If you are absent from one semester test for a valid reason, you will be required to write the special test towards the end of the semester. The scope of this test will be the combined scopes of the 1st and 2nd semester tests. The mark you attain for the special test will substitute the mark of the semester test you missed.

- If you are absent from the **exam**, the relevant faculty office should be informed without delay.

Sick notes (General Rules and Regulations A21 (p.8))

1. Only original medical certificates issued by persons and practitioners registered with the Health Professions Council of South Africa or the Allied Health Professions Council of South Africa will be accepted.
2. Medical certificates must contain the practice number, address, contact details and signature of the particular practitioner who issued the certificate. The certificate must clearly identify the student and must reflect that a consultation took place and/or that the student was examined on a specific date. The certificate must furthermore indicate the specific days during which the student is unfit to participate in academic activities.
3. A medical certificate will not be accepted if it merely states that the student appeared ill or declared himself or herself unfit.

Affidavits (General Rules and Regulations A22 (p.8))

In case of other valid reasons (apart from illness) for absence from tests, examinations, etc., an affidavit will only be accepted if accompanied by the relevant substantiating documentation, e.g., case report or criminal charge with case number obtained from

a police station, valid medical certificate for injuries, a death certificate for a funeral, etc.

Please note that submission of fraudulent sick notes and affidavits is a criminal offense, which will lead to disciplinary action and may result in dismissal.

3. Semester test dates

These will be announced on ClickUP.

4. Students with special needs

Students with special assessment needs (e.g., extra time) must contact the UP disability centre for a letter confirming this need. **Please follow all guidelines and instructions in this regard meticulously.** It is important to inform the module coordinator about any special needs. You need the original documentation at all assessment opportunities.

5. Marked tests and test queries

Semester tests: You will obtain the results of semester tests about 7-10 days after the test was written. The memos of the semester tests (or parts there-of) will be published on ClickUP. If you have a query related to the marking of a semester test, you will be given an opportunity to do so. This must be done within three (working) days after the tests have been handed out. The entire test could be re-marked in such a case.

Class tests: Queries related to the marking of the class tests must be raised during the following tutorial class. No changes will be made there-after.

6. Mark list

Please check your marks on the ClickUP Ultra Grade Centre regularly. When a mark is omitted or incorrect, you need to inform the course administrator as soon as possible. Changes will only be made within one week after the marks have been uploaded.

7. Semester mark

Your semester mark will be computed with the following weights:

Semester tests	70%
Class tests	30%

8. Entry requirement for the exam

In order to be admitted to the exam, you need a semester mark of at least 30% and you need to comply with the attendance requirements of the course.

9. Final mark

Your final mark will be computed with the following weights:

Semester mark	50%
Examination mark	50%

To pass this course a minimum of 50% is required for the final mark, as well as a subminimum of 40% in the exam.

10. Supplementary examination

A supplementary exam will only be awarded in one of the following cases:

- (a) The student has a final mark of between 40% and 49%;
- (b) The student has obtained a final mark of at least 50%, but has not met the sub-minimum of 40% for the exam.

Please note that the final mark for a supplementary exam does not take into account the semester mark and the maximum mark awarded is 50%.

11. Disciplinary cases

The policy of the Department is to refer *all* incidents in which there is a suspicion of irregularity to the Disciplinary Committee of the University.

12. M-numbers

You will be assigned a number (called an M-number) that you should remember and write on every class test, semester test and exam paper.

C. Course contents

1. Use of the study guide

In the study guide, the module is divided into a number of **THEMES** and each theme is further subdivided into **UNITS**. This should provide you with a good overview of the structure of the module. For each lecture unit, a textbook reference for the lecture unit is given under **Source**. In every lecture unit, there are **Learning objectives**. These spell out exactly what you should be able to do in that unit. In addition, you should be able to combine knowledge from different units to solve particular problems.

2. Module objectives

To have a thorough knowledge of functions, limits, continuity, derivatives and integration and to be able to apply these concepts in problem solving.

3. Module outcomes

After completion of this module a student will

- know the definition, as well as the properties of linear, power, trigonometric, exponential and logarithmic functions;
- be able to sketch linear, power, trigonometric, exponential and logarithmic functions (also using shifts and translations);
- be able to find limits of a function of a single variable and know what role limits play in finding derivatives;
- be able to identify continuous functions and differentiable functions;
- be able to find derivatives;
- be able to use derivatives to solve problems and to sketch graphs;
- be able to find anti-derivatives;
- be able to evaluate integrals;
- know how to add and subtract vectors, as well as how to find the dot and the cross product of two vectors in the three dimensional space.

4. Module structure

The subject matter for the course is divided into six themes:

- | | |
|---------------------------------------|-------------|
| 1. Functions | 15 lectures |
| 2. Limits, continuity and derivatives | 9 lectures |
| 3. Differentiation | 5 lectures |
| 4. Application of differentiation | 9 lectures |
| 5. Integration | 6 lectures |
| 6. Vector algebra | 5 lectures |

THEME 1: FUNCTIONS

Unit 1.1 Numbers, intervals and inequalities

Source: Stewart, Appendix A pp A2 - A6.

Learning objectives

On completion of this unit you should

1. know the definitions of the following number systems: the set $\mathbb{N}$ of all natural numbers, the set $\mathbb{Z}$ of all integers, the set $\mathbb{Q}$ of all rational numbers, the set $\mathbb{I}$ of all irrational numbers and the set $\mathbb{R}$ of all real numbers;
2. be able to use set notation;
3. be able to use the word "and" for the intersection of two sets;
4. be able to use the word "or" for the union of two sets;
5. be able to find the union/intersection of two sets;
6. be able to use interval notation;
7. know the rules for inequalities and be able to use it;
8. be able to solve inequalities.

Remarks

1. The set $\mathbb{N}$ of all natural numbers is the set $\mathbb{N} = \{1, 2, 3, \dots\}$.
2. Pay special attention to Table [1] on p A4. You must be able to describe an interval in interval notation and in set notation.
3. **Use a graph to solve inequalities where possible.** It is quicker and easier than the algebraic approach.
4. You must know and be able to use the following formula:

$$ax^2 + bx + c = 0 \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Assigned Problems

Stewart, Exercises A p A9: Numbers 17, 19, 25, 27, 29, 35, 39, 42, 58, 60.

Unit 1.2 The absolute value

Source: Stewart, Appendix A pp A6 - A9.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the absolute value.
2. be able to write down and use the properties of the absolute value.
3. be able to solve inequalities with absolute values.
4. be able to sketch the graph of $y = |f(x)|$.

Remarks

1. If $x \in \mathbb{R}$ then $\sqrt{x^2} = |x|$.
2. When solving inequalities with absolute values, use the theorems in [6] on p A7.
3. Omit the triangle inequality on p A8 and Example 9 on p A9.
4. You can also use the following theorem: If x and y are real numbers, then $|x| \leq |y|$ if and only if $x^2 \leq y^2$.

Assigned Problems

1. Stewart, Exercises A p A9: Numbers 1, 3, 7, 9, 12, 45, 47, 49, 51, 53, 55, 56.
 2. Sketch the graphs of:
 - a. $y = |x - 3|$
 - b. $y = |x^2 - 16|$
 - c. $y = |3x - 2|$
-

Unit 1.3 What is a function?

Source: Stewart, Section 1.1 pp 8 - 21.

Learning objectives

On completion of this unit you should

1. be able to identify a function.
2. be familiar with the four possible ways of representing a function.
3. be able to find the domain of a function.
4. be able to find the range of some functions by using the graph of the function.

5. know how to test whether a curve in the xy -plane is the graph of a function by using the Vertical Line Test.
6. be familiar with piecewise-defined functions.
7. be able to sketch the function $y = |f(x)|$ if the graph of the function $y = f(x)$ is given.
8. be able to write down and use the definition of an even function.
9. be able to write down and use the definition of an odd function.
10. be able to write down and use the definition of an increasing function.
11. be able to write down and use the definition of a decreasing function.

Assigned Problems

Stewart, Exercises 1.1 p 17: Numbers 1, 3, 7, 9, 25, 31, 33, 35, 41, 42, 45, 47, 51, 55, 71.

Unit 1.4 Radian measure and trigonometric functions

Source: Stewart, Appendix D pp A24 - A35.

Learning objectives

On completion of this unit you should

1. be able to use radian measure.
2. be able to write down and use the definitions of the six trigonometric functions.
3. be able to sketch the graphs of the functions $y = \sin x$, $y = \cos x$ and $y = \tan x$.
4. be able to write down and use some trigonometric identities.
5. be able to solve trigonometric equalities and inequalities.

Remarks

1. Omit Example 5.
2. Memorize identities 7, 8, 13, 14, 16 and 17 on pp A28 - A29. The others can be easily deduced from these, make sure you can do this.
3. You must be able to calculate all the function values in the table on p A27.
4. You must be able to find the exact trigonometric function values for the angles π , $\frac{\pi}{2}$, $\frac{\pi}{3}$, $\frac{\pi}{4}$, $\frac{\pi}{6}$ and any related angles without using a calculator.
5. **We always use radian measure when we work with trigonometric functions.**

Assigned Problems

1. Stewart, Exercises D p A33: Numbers 3, 9, 19, 20, 25, 26, 31, 32, 34, 45, 47, 59, 69, 71, 72, 75, 76.
 2. Find $\sin(75\pi)$ and $\cos\left(\frac{17\pi}{4}\right)$ without using a calculator.
-

Unit 1.5 A catalogue of essential functions - Self study

Source: Stewart, Section 1.2 pp 21 - 36.

Learning objectives

On completion of this unit you should

1. be able to identify the following functions: linear functions, power functions, polynomials, algebraic functions, rational functions, exponential functions and logarithmic functions.
2. be able to sketch the graphs of linear functions, power functions, some polynomials, exponential functions and logarithmic functions.

Remark: Omit Examples 2, 3 and 4.

Assigned Problems

Stewart, Exercises 1.2 p 33: Numbers 1, 3, 4, 6, 8, 11, 13, 14.

Unit 1.6 New functions from old functions

Source: Stewart, Section 1.3 pp 36 - 45.

Learning objectives

On completion of this unit you should

1. be able to transform a function through translations: vertical and horizontal shifts, stretching, compressing and reflecting.
2. be able to combine two functions f and g to form new functions $f + g$, $f - g$, fg and $\frac{f}{g}$ and be able to determine the domains of these functions.
3. be able to form the composite functions $f \circ g$ and $g \circ f$ for given functions f and g and be able to determine the domain of the new functions.
4. be able to decide what the functions f and g are if the composite function $f \circ g$ is given.

Remarks

1. You should be able to sketch the new transformed function by using the graph(s) of the old function(s).
2. Omit Example 4 on p 39.
3. The functions $y = \sin(Bx)$ and $y = \cos(Bx)$ have period $\frac{2\pi}{|B|}$.

Assigned Problems

1. Stewart, Exercises 1.3 p 42: Numbers 1, 3, 6, 7, 13, 14, 33, 36, 37, 43, 45, 47, 48, 51, 57, 65.
2. Stewart, Exercises D p A35: Numbers 83 and 87.
3. Sketch the following functions on the interval $[-2\pi, 2\pi]$:
 - a. $f(x) = 2 \sin x$
 - b. $g(x) = \sin(2x)$
 - c. $h(x) = 2 - \sin x$

Unit 1.7 Exponential functions

Source: Stewart, Section 1.4 pp 45 - 53.

Learning objectives

On completion of this unit you should

1. be able to write down and use the equation of an exponential function.
2. know that e is the symbol for a significant irrational number.
3. be able to sketch the graphs of the following exponential functions: $y = a^x, a > 1$ and $y = a^x, 0 < a < 1$ and $y = e^x$ and $y = e^{-x}$.
4. be able to sketch the graphs of shifted and translated exponential functions.
5. be able to write down and use the laws of exponents.

Remarks

1. You do not have to know how a^x , with x an irrational number, is defined (if interested, see the explanation on pp 50 - 51).
2. Omit Example 6 on p 51.

Assigned Problems

Stewart, Exercises 1.4 p 52: Numbers 1, 3, 13, 15, 16, 19, 21, 29(a) to (c).

Unit 1.8 Inverse functions

Source: Stewart, Section 1.5 pp 54 - 57.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a one-to-one function.
2. know how to test whether a function is one-to-one by using the Horizontal Line Test.
3. be able to write down and use the definition of an inverse function f^{-1} .
4. be able to write down the relationship between the domains and ranges of f and f^{-1} .
5. be able to write down and use the cancellation equations of the composites of f and f^{-1} .
6. be able to find the inverse function of a one-to-one function f .
7. be able to explain the relationship between the graphs of f and f^{-1} .

Remarks

1. The definition of a one-to-one function can be written in two ways:
A function $f : A \rightarrow B$ is one-to-one if, for all $x_1, x_2 \in A$, $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$.
We can also say: A function $f : A \rightarrow B$ is one-to-one if $f(x_1) = f(x_2) \implies x_1 = x_2$.
2. The three steps to find a inverse function is given on p 56.

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 5, 7, 11, 17, 20, 25.

Unit 1.9 Logarithmic functions

Source: Stewart, Section 1.5 pp 57 - 61.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the logarithmic function, $y = \log_a x$.
2. be able to write down and use the laws of logarithms.
3. be able to write down and use the definition of the natural logarithmic function, $y = \ln x$.
4. be able to write down and use the properties of the natural logarithm.
5. be able to use the cancellation equations of the composites of $y = \ln x$ and $y = e^x$ to solve equations and inequalities.
6. be able to sketch the graph of the function $y = \ln x$.

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 27, 28, 30, 40, 42, 57, 59, 62, 63, 64.

Unit 1.10 Inverse trigonometric functions

Source: Stewart, Section 1.5 pp 61 - 64.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the inverse trigonometric functions $y = \arcsin x$, $y = \arccos x$ and $y = \arctan x$.
2. be able to write down and use the properties of the inverse trigonometric functions $y = \arcsin x$, $y = \arccos x$ and $y = \arctan x$.
3. be able to sketch the graphs of the functions $y = \arcsin x$, $y = \arccos x$, $y = \arctan x$.
4. be able to use the cancellation equations of the composites of f and f^{-1} to solve equations.

Remark

Omit the definitions of the other three inverse trigonometric function ([12] on p 64).

Assigned Problems

Stewart, Exercises 1.5 p 64: Numbers 69, 70, 71, 72, 74, 75, 76, 77, 78, 81.

THEME 2: LIMITS, CONTINUITY AND DERIVATIVES

Unit 2.1 The tangent and velocity problems

Source: Stewart, Section 2.1 pp 78 - 82.

Learning objectives

On completion of this unit you should

1. be able to estimate the slope of a tangent line to a curve by calculation of the limit of the slopes of secant lines.
2. be able to explain the concepts "average velocity over an interval" and "instantaneous velocity at time t ".
3. be able to estimate the instantaneous velocity at time t by finding the average velocity over intervals if a distance function is given.

Remark

At a later stage we will develop easier ways of finding instantaneous velocity and the slopes of tangent lines.

Assigned Problems

Stewart, Exercises 2.1 p 82: Numbers 1(a), 1(b), 8.

Unit 2.2 The limit of a function

Source: Stewart, Section 2.2 pp 83 - 94.

Learning objectives

On completion of this unit you should

1. be able to find the limit of a function at a point if a graph is given.
2. be able to explain what is meant by one-sided limits and be able to interpret these ideas graphically.
3. be able to find $\lim_{x \rightarrow a} f(x)$ by using the one-sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ (when necessary).
4. be able to explain the informal definition of an infinite limit ($\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$) and be able to interpret this idea graphically.
5. be able to find vertical asymptotes of a function.

Remarks

1. It is important to understand the informal idea of a limit as given on p 83. The formal precise definition is given on p 106. You do not have to know this definition.

- Examples 3 and 5 are important, they show why we need better methods to find limits.
- In order to find the vertical asymptotes of a function: Find the domain of the function. If $x = c$ is in the domain of the function then $x = c$ is NOT a vertical asymptote. If the domain is (a, ∞) , calculate $\lim_{x \rightarrow a^+} f(x)$. If the domain is $(-\infty, a)$, calculate $\lim_{x \rightarrow a^-} f(x)$. If the domain is (a, b) , calculate $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$.

Assigned Problems

Stewart, Exercises 2.2 p 92: Numbers 1, 2, 5, 7, 9, 11, 15, 17, 29, 32, 33, 35, 36.

Unit 2.3 Limit laws

Source: Stewart, Section 2.3 pp 94 - 104.

Learning objectives

On completion of this unit you should

- be able to use the appropriate limit law to find
 - the limit of the sum of two functions;
 - the limit of the difference of two functions;
 - the limit of a constant multiple of a function;
 - the limit of the product of two functions;
 - the limit of the quotient of two functions;
- be able to use the direct substitution property to find the limit of polynomials and rational functions;
- be able to determine whether a limit exists by looking at the one-sided limits at the point;
- be able to write down and use the squeeze theorem;
- be able to use the limit $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, to determine similar limits, e.g., $\lim_{x \rightarrow 0} \frac{x}{\sin x}$, $\lim_{x \rightarrow 0} \frac{\tan x}{x}$ and $\lim_{x \rightarrow 0} \frac{x}{\tan x}$.

Remarks

- These laws can be proved by using the formal definition of a limit.
- The method used in Example 6 is important.
- Omit Example 10 on p 101.

4. The “proof” that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ appears on p 195 but you do not have to know it.

Assigned Problems

1. Stewart, Exercises 2.3 p 102: Numbers 2, 13, 15, 21, 23, 27, 29, 41, 46, 47, 52, 61, 64.
2. Stewart, Exercises 3.3 p 197: Numbers 45, 46, 53, 54, 55.

Unit 2.4 Continuity

Source: Stewart, Section 2.5 pp 115 - 126.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of continuity of a function at a point;
2. be able to explain the definition of continuity at a point graphically;
3. be able to determine if a given function is continuous at a point;
4. be able to distinguish between removable, jump and infinite discontinuities;
5. be able to write down and use the definitions of “continuous from the left” and “continuous from the right”;
6. be able to test whether a function is continuous over an interval. You have to distinguish between the intervals $[a, b]$, (a, b) , $(a, b]$, $[a, b)$, etc.;
7. be able to use the theorems on the continuity of combinations of functions (sum, difference, constant multiple, products, quotient);
8. know that the following functions are continuous in every point of their domains
 - polynomials · rational functions · root functions
 - exponential functions · logarithmic functions · trigonometric functions
 - inverse trigonometric functions
9. know that the composite of continuous functions is continuous and be able to use this fact to calculate limits.

Remarks

1. Omit the proofs of the theorems and the Intermediate Value Theorem.
2. Omit Examples 2(d), 3 and 10.

Assigned Problems

Stewart, Exercises 2.5 p 124: Numbers 3, 5, 7, 9, 13, 20, 23, 41, 43, 48, 49.

Unit 2.5 Limits at infinity and horizontal asymptotes

Source: Stewart, Section 2.6 pp 127 - 140.

Learning objectives

On completion of this unit you should

1. be able to explain what is meant by the following limits:

$$\lim_{x \rightarrow \infty} f(x) = L, \quad \lim_{x \rightarrow -\infty} f(x) = L, \quad \lim_{x \rightarrow \infty} f(x) = \infty, \quad \lim_{x \rightarrow -\infty} f(x) = \infty,$$

$$\lim_{x \rightarrow \infty} f(x) = -\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

and to interpret the limits graphically.

2. be able to find horizontal asymptotes of a function.
3. be able to find $\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)}$, with $f(x)$ and $g(x)$ polynomials by using the fact that $\lim_{x \rightarrow \pm\infty} \frac{1}{x^r} = 0$, for r a positive rational number.

Remarks

1. You have to memorize the two limits given in [4] on p 132.
2. Example 4 shows that there may be more than one horizontal asymptote.
3. Omit Example 12. We will sketch functions later.
4. Omit the precise definitions and examples on pp 134 - 137.

Assigned Problems

Stewart, Exercises 2.6 p 137: Numbers 3, 5, 8, 15, 17, 19, 23, 27, 33, 37, 41, 47, 49, 51, 52, 56.

Unit 2.6 Derivatives and rates of change

Source: Stewart, Section 2.7 pp 140 - 152.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the tangent line to a curve.
2. be able to find the equation of the tangent line to a graph.
3. be able to interpret average velocity and instantaneous velocity as slopes of graphs.

4. be able to find the (instantaneous) velocity of an object if the distance is given as a function of the time.
5. be able to write down and use the definition of the derivative of a function at a point.
6. know that the derivative is the slope of the tangent line to the curve.
7. be able to find derivatives of functions by using the definition.
8. be able to estimate derivatives from tabled data.
9. be able to distinguish between average rate of change over an interval and (instantaneous) rate of change at a point.

Remarks

1. There are two formulas to calculate the derivative at a point, [4] and [5] on p 144. You have to know both.
2. Omit Examples 7 and 8.

Assigned Problems

Exercises 2.7 p 149: Numbers 3, 7, 12, 16, 17, 18, 19, 28, 34, 39, 43, 45, 57.

Unit 2.7 The derivative as a function

Source: Stewart, Section 2.8 pp 153 - 166.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the derivative function;
2. be able to find the formula of the derivative function by using the definition;
3. be able to sketch the graph of the derivative function if the graph of the function is given;
4. know the other notations for the derivative;
5. be able to write down and use the definition of a function that is differentiable at a point;
6. be able to write down and use the definition of a function that is differentiable on an interval;
7. know and be able to prove that a differentiable function is continuous in a point and that the converse of the theorem is not true;

8. know the three ways for a function not to be differentiable at a point;
9. be able to illustrate graphically when a function is differentiable in a point;
10. be able to illustrate graphically when a function is not differentiable in a point;
11. know the notation for higher derivatives.

Remarks

1. You have to be able to prove Theorem 4 on p 157.
2. Omit Examples 6 and 7.

Assigned Problems

Stewart, Exercises 2.8 p 161: Numbers 3, 4, 9, 23, 27, 41, 42, 47, 49, 51, 63.

THEME 3: DIFFERENTIATION

Unit 3.1 Derivatives of polynomials and exponential functions

Source: Stewart, Section 3.1 pp 174 - 184.

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of a constant function, power functions, the natural exponential function, constant multiple of a function, sum of functions, difference of functions;
2. be able to use the rules and formulas in problems.

Remark

You do not have to prove any of the rules in this unit.

Assigned Problems

Stewart, Exercises 3.1 p 181: All odd numbers from 3 to 35, 50, 53, 59, 60, 61, 62, 65, 67, 71, 78, 85.

Unit 3.2 The product rule and the quotient rule

Source: Stewart, Section 3.2 pp 185 - 191.

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of the product of functions;
2. be able to write down and use the rule to find the derivative of the quotient of functions;
3. be able to use the rules in problems.

Assigned Problems

Stewart, Exercises 3.2 p 189: All odd numbers from 3 to 25, 33, 39(a), 43, 45, 53, 55, 66.

Unit 3.3 Derivatives of trigonometric functions

Source: Stewart, Section 3.3 pp 191 - 199.

Learning objectives

On completion of this unit you should

1. be able to write down the derivatives of the six trigonometric functions;
2. be able to use these derivatives in problems.

Remarks

1. You should be able to prove that $\frac{d}{dx} \sin x = \cos x$ and that $\frac{d}{dx} \cos x = -\sin x$ by using the limit definition.

You may assume that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, and do not have to know the proof on p 195.

You must be able to prove that $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$ (p 196).

2. You should be able to prove the formulas for the derivatives of the other four trigonometric functions by using the derivatives of sin and cos and the quotient rule.

Assigned Problems

Exercises 3.3 p 197: All odd numbers from 9 to 27, 32(a), 34(a), 39, 41, 43.

Unit 3.4 The chain rule

Source: Stewart, Section 3.4 pp 199 - 209.

Learning objectives

On completion of this unit you should

1. be able to use the chain rule to find the derivative of a composite function;
2. be able to write down the derivative of any exponential function $f(x) = a^x$.

Remark: You do not have to prove the chain rule.

Assigned Problems

Stewart, Exercises 3.4 p 206: 3, all odd numbers from 7 to 21 and 39 to 51, 53, 58, 59, 65, 68, 69, 85, 90 a,b.

Unit 3.5 Implicit differentiation

Source: Stewart, Section 3.5 pp 209 - 216, 222 - 223.

Learning objectives

On completion of this unit you should

1. know when a function is defined implicitly;
2. be able to differentiate a function given implicitly by using implicit differentiation;
3. be able to use the technique and formulas in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of the functions $y = \arcsin x$ and $y = \arctan x$, see pages 222-223.
2. You do not have to know the derivatives of the other inverse trigonometric functions (p 223).

Assigned Problems

Stewart, Exercises 3.5 p 214: All odd numbers from 5 to 21, 27, 31, 43, 47, 61, 64 and Exercises 3.6 p. 224: 63, 65, 69, 75.

Unit 3.6 Derivatives of logarithmic functions and inverse trigonometric functions

Source: Stewart, Section 3.6 pp 217 - 225.

Learning objectives

On completion of this unit you should

1. be able to write down and use the formula for the derivative of the logarithmic function $y = \log_a x$;
2. be able to write down and use the formula for the derivative of the logarithmic functions $y = \ln x$ and $y = \ln |x|$;
3. know when and how to use logarithmic differentiation; item be able to write down the formulas for the derivatives of the two inverse trigonometric functions $y = \arcsin x$ and $y = \arctan x$;
4. be able to use the formula and technique in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of the logarithmic functions (pp 217-218).

2. The four cases given on p 221 are very important.
3. The following result is worth remembering:

$$e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

4. You must be able to prove the formulas for the derivatives of the functions $y = \arcsin x$ and $y = \arctan x$.
5. You do not have to know the derivatives of the other inverse trigonometric functions (p 223).

Assigned Problems

Exercises 3.6 p 224: All odd numbers from 7 to 25, 32, 35, 39, 45, 51, 57.

Unit 3.7 Hyperbolic functions

Source: Stewart, Section 3.11 pp 261 - 268.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
2. be able to sketch the graphs of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
3. be able to write down and use the derivatives of the functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$;
4. be able to use the formulas and graphs in problems.

Remarks

1. You should be able to prove the derivatives of the three hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
2. You do not have to memorize the identities on p 261. The identities can be proven by using the definitions of the two hyperbolic functions $\sinh$ and $\cosh$.
3. Omit the section on inverse hyperbolic functions.

Assigned Problems

Stewart, Exercises 3.11 p 266: Numbers 1, 2, 3, 27, 36, 37, 39.

THEME 4: APPLICATIONS OF DIFFERENTIATION

Unit 4.1 Maximum and minimum values

Source: Stewart, Section 4.1 pp 276 - 287.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an absolute maximum of a function, an absolute minimum of a function, a local maximum of a function, a local minimum of a function and a critical number of a function;
2. be able to write down and use the Extreme Value Theorem, Fermat's Theorem;
3. know that the converse of Fermat's Theorem is false;
4. be able to find the critical numbers of a function;
5. be able to use the Closed Interval Method to find the absolute extremes of a continuous function on a closed interval;
6. be able to find the absolute and local extremes of a function if the graph of the function is given.

Remarks

1. You do not have to prove any theorems in this unit.
2. It is important to remember that if c is a critical number it does not mean that $f(c)$ is a local extreme. You have to use the tests given in Unit 4.3 to test whether it is a local extreme or not.
3. Omit Example 9.

Assigned Problems

Stewart, Exercises 4.1 p 286: Numbers 3, 5, 9, 19, 20, 21, 27, 31, 33, 41, 45, 55, 59, 61, 63, 64, 65.

Unit 4.2 The Mean Value Theorem

Source: Stewart, Section 4.2 pp 290 - 296.

Learning objectives

On completion of this unit you should

1. be able to write down Rolle's Theorem and be able to apply it in problems;
2. be able to write down the Mean Value Theorem and be able to apply it in problems.

Remarks

1. You do not have to prove any of the theorems in this unit.
2. Omit Example 6.

Assigned Problems

Stewart, Exercises 4.2 p 295: Numbers 3, 6, 9, 12, 17, 21, 31, 33.

Unit 4.3 How the derivative affects the shape of the graph

Source: Stewart, Section 4.3 pp 296 - 309.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a function that is increasing on an interval;
2. be able to write down and use the definition of a function that is decreasing on an interval;
3. be able to use the Increasing/Decreasing Test to find the interval(s) on which a function is increasing and the interval(s) on which a function is decreasing;
4. be able to use the First Derivative Test to find the local extremes of a function;
5. be able to write down and use the definition of a function that is concave upward on an interval;
6. be able to write down and use the definition of a function that is concave downward on an interval;
7. be able to use the Concavity Test to find the interval(s) on which a function is concave downward and the interval(s) on which a function is concave upward;
8. be able to write down and use the definition of an inflection point;
9. be able to use the second derivative to find the inflection point(s) of a function;
10. be able to use the Second Derivative Test to find the local extremes of a function.

Remarks

1. The definition of an increasing / a decreasing function is on p 17.
2. Leave Examples 6, 7 and 8 for now. We will do curve sketching in Unit 4.5.

Assigned Problems

Stewart, Exercises 4.3 p 305: Numbers 1, 5, 7, 9, 15, 16, 19, 25, 26, 29, 40, 53, 57, 84.

Unit 4.4 Indeterminate forms and L'Hospital's rule

Source: Stewart, Section 4.4 pp 309 - 319.

Learning objectives

On completion of this unit you should

1. be able to explain what indeterminate limit forms are;
2. be able to write down the seven different types of indeterminate forms;
3. be able to identify when a limit is of indeterminate form;
4. be able to use the rule of L'Hospital to find certain limits of indeterminate forms.

Remark

You always have to write down when a limit has an indeterminate form and you have to indicate that you use L'Hospital's rule, e.g.,

$$\begin{aligned} & \lim_{x \rightarrow \infty} \frac{e^x}{x^2} \quad \left(\text{form } \frac{\infty}{\infty}\right) \quad \text{L'Hospital} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2x} \quad \left(\text{form } \frac{\infty}{\infty}\right) \quad \text{L'Hospital} \\ &= \lim_{x \rightarrow \infty} \frac{e^x}{2} \\ &= \infty. \end{aligned}$$

Assigned Problems

Stewart, Exercises 4.4 p 316: Numbers 7, 9, 11, 15, 17, 25, 29, 49, 51, 53, 56, 57, 58, 59, 60, 65, 77.

Unit 4.5 Curve sketching

Source: Stewart, Section 4.5 pp 320 - 329.

Learning objectives

On completion of this unit you should be able to sketch the graph of a given function.

Remark

To sketch the graph of a function, you have to follow the guidelines given in this section, i.e., you have to

1. find the domain of the function;

2. if possible, find the intercepts of the function with the x -axis and with the y -axis;
3. determine whether the function is even, odd or periodic, or none of these;
4. find the horizontal asymptotes (if any) of the function;
5. find the vertical asymptotes (if any) of the function;
6. find the interval(s) on which the function is increasing and the interval(s) on which the function is decreasing;
7. find the local extremes of the function (use number 6);
8. find the interval(s) on which the function is concave upward and the interval(s) on which the function is concave downward;
9. find the inflection points of the function;
10. use the information above to sketch the graph of the function.

Remark: Do examples 6, 7 and 8 on pp 302 - 305.

Assigned Problems

1. Stewart, Exercises 4.5 p 327: Numbers 7, 13, 27, 41, 52, 53.
2. Stewart. Exercises 4.3 p 305: Numbers 35, 37, 38.

Unit 4.6 Optimization problems - Self study

Source: Stewart, Section 4.7 pp 336 - 349.

Learning objectives

On completion of this unit you should

1. know what is meant by an optimization problem;
2. be able to convert a word problem into a mathematical optimization problem by setting up the function that has to be minimized or maximized;
3. be able to use the methods in this theme to find absolute extreme values;
4. be able to solve optimization problems such as described in the examples.

Remarks

1. The main task in this section is to convert a word problem into a mathematically formulated problem.
2. Making a sketch is invaluable.

Assigned Problems

Stewart, Exercises 4.7 p 342: Numbers 1, 3, 5, 9, 21.

THEME 5: INTEGRATION

Unit 5.1 Antiderivatives

Source: Stewart, Section 4.9 pp 356 - 364.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an antiderivative of a function;
2. be able to find the most general antiderivative of a function;
3. be able to find a function if the derivative(s) of the function is(are) given.

Remarks

1. You can omit the section about linear motion (pp 360 - 361).
2. Memorize the table on p 358.

Assigned Problems

Stewart, Exercises 4.9 p 361: All the odd numbers from 5 to 25, 31, 39, 45, 47, 55, 65, 67.

Unit 5.2 The area problem

Source: Stewart, Section 5.1 pp 372 - 384.

Learning objectives

On completion of this unit you should

1. be able to estimate the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$ by using the sum of the areas of rectangles. The rectangles are constructed by using as sample points either the left endpoints or the right endpoints of each subinterval;
2. be able to find an upper and lower estimate of the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$. The function must be either increasing or decreasing on the interval.

Remark: Omit Example 2

Assigned Problems

Stewart, Exercises 5.1 p 381: Numbers 1, 4, 7, 9.

Unit 5.3 The definite integral

Source: Stewart, Section 5.2 pp 384, 385, 391 and 393.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a definite integral;
2. know what the integrand and limits of integration of a definite integral are;
3. be able to use a Riemann sum to approximate a definite integral;
4. know and be able to use the properties of a definite integral;
5. be able to evaluate integrals by interpreting it in terms of areas;
6. know the comparison properties of the integral.

Remarks

1. Study only Examples 1,5 and 8.
2. If f is integrable on an interval $[a, b]$ and $f(x) \geq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \geq 0$. In fact, $\int_a^b f(x)dx$ is equal to the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.
3. If f is integrable on an interval $[a, b]$ and $f(x) \leq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \leq 0$. In fact, $\int_a^b f(x)dx$ is equal to the negative of the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

Assigned Problems

Stewart, Exercises 5.2 p 394: Numbers 5(a&b), 7, 35, 41, 43, 45, 59, 60, 67.

Unit 5.4 The Fundamental Theorem of Calculus

Source: Stewart, Section 5.3 pp 402 - 409.

Learning objectives

On completion of this unit you should know and be able to proof and use the Fundamental Theorem of Calculus, part 2 (p 403).

Remarks

1. The following proof will be discussed in class:

Theorem 1. *The Fundamental Theorem of Calculus.*

If f is continuous on $[a, b]$, then $\int_a^b f(x)dx = F(b) - F(a)$, where F is any antiderivative of f , that is, a function such that $F' = f$.

Proof. Divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$. Let $a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$ be the endpoints of these intervals and let F be an antiderivative of f , that is, $F' = f$. This means

- F is differentiable on $[a, b]$;
- F is continuous on $[a, b]$ (from Theorem 4 on p 157).

We will now apply the Mean Value Theorem to each of the subintervals $[x_{i-1}, x_i], i = 1, 2, \dots, n$. Since

- F is continuous on $[x_{i-1}, x_i]$;
- F is differentiable on (x_{i-1}, x_i) ,

there exists a number c_i in the interval (x_{i-1}, x_i) such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) = f(c_i) \quad (\text{it is given that } F' = f). \quad (1)$$

Let this number c_i be the sample point in each interval. As f is continuous on $[a, b]$, f is integrable on $[a, b]$ (Theorem 3, p 386) and

$$\begin{aligned} \int_a^b f(x)dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i)\Delta x, \text{ definition of definite integral (p.384)} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \Delta x, \text{ from (1)} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} (x_i - x_{i-1}), \text{ as } \Delta x \text{ is the length of each subinterval} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n F(x_i) - F(x_{i-1}) \\ &= \lim_{n \rightarrow \infty} (F(x_1) - F(x_0)) + (F(x_2) - F(x_1)) + \dots + (F(x_n) - F(x_{n-1})) \\ &= \lim_{n \rightarrow \infty} (F(x_n) - F(x_0)) \\ &= F(b) - F(a). \end{aligned}$$

□

2. Work through Examples 5 to 9.

Assigned Problems

Stewart, Exercises 5.3 p 406: 21, 22, All the odd numbers between 25 and 55, 63.

Unit 5.5 Indefinite integrals

Source: Stewart, Section 5.4 pp 409 - 412.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an indefinite integral;
2. be able to evaluate indefinite integrals.

Assigned Problems

1. Stewart, Exercises 5.4 p 415: Number 1, all the odd numbers from 7 to 21, 39, 42, 48, 53.
 2. Stewart, Exercises 5.5 p 425: All the odd numbers from 1 to 53 and from 59 to 73.
Integrate without making a substitution.
-

THEME 6: VECTOR ALGEBRA

Unit 6.1 Three-dimensional coordinate systems

Source: Stewart, Section 12.1 pp 830 - 836.

Learning objectives

On completion of this unit you should

1. be able to explain what a three-dimensional rectangular coordinate system is;
2. be able to calculate the distance between two points in the three-dimensional space;
3. be able to find the projections of a given point on coordinate planes or coordinate axes;
4. be able to write down the equation of a sphere with given centre and radius.

Assigned Problems

Stewart, Exercises 12.1 p 835: Numbers 3, 4, 11, 12, 15, 19.

Unit 6.2 Vectors

Source: Stewart, Section 12.2 pp 836 - 846.

Learning objectives

On completion of this unit you should

1. know what a vector is;
2. be able to represent vectors geometrically;
3. understand that a vector has a unique direction and length but not a unique position;
4. be able to represent vectors algebraically;
5. be able to write down a vector with given initial and end points;

6. be able to find the length of a vector;
7. be able to write down and use the definitions of vector addition and scalar multiplication;
8. be able to write down and use the properties of vectors;
9. be able to write down and use the definition of parallel vectors;
10. be familiar with the standard basis vectors in two- and three-dimensional space and be able to express any vector in terms of these;
11. know what is meant by a unit vector;
12. be able to find unit vectors in the same or opposite direction of a given non-zero vector.

Assigned Problems

Stewart, Exercises 12.2 p 843: Numbers 3, 4, 7, 9, 15, 19, 21, 22, 23, 26, 29.

Unit 6.3 The dot product

Source: Stewart, Section 12.3, pp 847 - 854.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the dot product;
2. be able to determine the dot product of given vectors;
3. be able to write down and use the properties of the dot product;
4. be able to use the dot product to find the angle between two given vectors;
5. be able to use the dot product to determine whether two vectors are perpendicular (orthogonal);
6. be able to determine directional angles and cosines;
7. be able to find scalar and vector projections.

Remark: You do not have to do any proofs.

Assigned Problems

Stewart, Exercises 12.3 p 852: Numbers 1, 4, 7, 11, 12, 15, 19, 21, 23, 25, 27, 29, 33, 38, 39, 43, 45, 46.

Unit 6.4 The cross product

Source: Stewart, Section 12.4 pp 855 - 863.

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the cross product of two vectors;
2. be able to find the cross product of two given vectors;
3. be able to find a vector orthogonal to two vectors;
4. be able to determine if two vectors are parallel using the cross product;
5. be able to write down and use the properties of the cross product;
6. know how to find the area of a parallelogram using the cross product.

Remarks

1. Cross products are only defined for three-dimensional vectors.
2. Note that the order of the vectors when calculating the cross product is important.
3. The introduction of a determinant is just to make it easier to calculate the cross product.
4. You do not have to know any proofs.

Assigned Problems

Stewart, Exercises 12.4 p 861: Numbers 1, 3, 7, 13, 19, 20, 27, 43.

APPENDIX: TECHNIQUE MASTERING EXERCISES

A - ABSOLUTE VALUE (Unit 1.1)

- Write in the form $\begin{cases} \dots & \text{if } x \geq \dots \\ \dots & \text{if } x < \dots \end{cases}$.
a. $|x - 2|$ **b.** $|3 + 6x|$ **c.** $|x^2 - 1|$ **d.** $|x^4 + 4|$
- Solve, using the relevant theorems, and write your answers in interval notation (where possible).
a. $|x - 2| = 0$ **b.** $|x - 2| > 0$ **c.** $|x - 2| < 0$ **d.** $|x - 2| < 1$
e. $|2x| < 7$ **f.** $|2x - 4| > 8$ **g.** $|x - 3| \leq 6$ **h.** $|2x + 5| \geq 6$
i. $|1 - 2x| = 3$ **j.** $|3 - 7x| \leq 6$ **k.** $|5 - 3x| = -3$ **l.** $|-5x - 8| > 12$
m. $|x| = |-3|$ **n.** $|3x| = |2x + 1|$ **o.** $\frac{x - 1}{|2x|} \leq 0$ **p.** $\frac{|2x|}{x - 1} \leq 0$
- Sketch the graphs of the following pairs of functions.
a. $f(x) = x + 4$ and $g(x) = |x + 4|$ **b.** $f(x) = 1 - x^2$ and $g(x) = |1 - x^2|$
c. $f(x) = -x^3$ and $g(x) = |-x^3|$ **d.** $f(x) = 2^x$ and $g(x) = |2^x|$

B - INEQUALITIES

- Solve and write your answers in interval notation.
a. $\frac{3 - x}{-4} \geq 0$ **b.** $\frac{-4}{3 - x} < 0$ **c.** $\frac{2x - 6}{3x + 9} \geq 0$
d. $\frac{x - 1}{x} < 4$ **e.** $\frac{|-4x|}{3 - x} < 0$ **f.** $\frac{2x - 6}{|3x + 9|} \geq 0$
- Use graphs to solve the inequalities and write your answers in interval notation.
a. $x^2 + 4 > 5$ **b.** $x^2 - 4 > 2 - x$
♣c. $x^6 \geq x^4$ **♣d.** $x^5 < x^7$

♣ - after you have done Unit 1.5

C - FUNCTIONS

Problems with ♣ - do after you have done Unit 1.4 and ♦ - do after you have done Units 1.7 and 1.9.

- Determine the domain of f . Calculate the function value $f(a)$ of f at the given point a .

a. $f(x) = \frac{1}{\sqrt{x-1}}, a = 5$ **b.** $f(x) = \frac{3}{|x-4|}, a = 1$ **c.** $f(x) = x^4 + x^3 + 1, a = 3$
d. $f(x) = \frac{1}{x^2 + x + 1}, a = 0$ **♣e.** $f(x) = \frac{1}{\sin x}, a = \frac{5\pi}{2}$ **♦f.** $f(x) = e^{\frac{1}{x}}, a = \ln 3$
g. $f(x) = \frac{\sqrt{x} + 1}{\sqrt{x} - 1}, a = 2$ **♣h.** $f(x) = \tan x, a = -\frac{\pi}{4}$ **i.** $f(x) = \frac{|x|}{x}, a = -1$
j. $f(x) = \frac{x^2 + 2x + 1}{x^2 - 1}, a = 0$ **♦k.** $f(x) = \frac{\ln x}{x - 1}, a = e$ **♦l.** $f(x) = \ln(1 + e^x), a = 0$

2. For the given functions f and g , determine $f + g, f - g, fg, \frac{f}{g}, f \circ g$ and $g \circ f$.

a. $f(x) = 2x + 5, g(x) = x^2$ **b.** $f(x) = x^2, g(x) = \frac{1}{x-1}$
♦c. $f(x) = e^x, g(x) = \ln x$ **♣d.** $f(x) = \cos x, g(x) = \frac{1}{x}$
e. $f(x) = \sqrt{x}, g(x) = x - 1$ **f.** $f(x) = \sqrt{x}, g(x) = 2^x$

D - TRIGONOMETRIC FUNCTIONS

1. Find, without using a calculator.

a. $\sin \frac{7\pi}{4}$ **b.** $\operatorname{cosec} \left(-\frac{7\pi}{4} \right)$ **c.** $\sin(35\pi)$
d. $\cos \left(-\frac{5\pi}{3} \right)$ **e.** $\sec \frac{5\pi}{3}$ **f.** $\sec \frac{15\pi}{2}$
g. $\tan \frac{7\pi}{6}$ **h.** $\cot \left(-\frac{7\pi}{6} \right)$ **i.** $\cos \left(-\frac{15\pi}{3} \right)$

2. Find, without using a calculator.

a. $\cos \theta, \tan \theta, \operatorname{cosec} \theta$ and $\cot \theta$ if $\sin \theta = \frac{7}{25}, \theta \in \left(\frac{\pi}{2}, \pi \right)$
b. $\sin \theta, \tan \theta, \sec \theta$ and $\operatorname{cosec} \theta$ if $\cos \theta = -\frac{4}{5}, \theta \in \left(\frac{\pi}{2}, \pi \right)$
c. $\sin \theta, \cos \theta, \cot \theta$ and $\sec \theta$ if $\tan \theta = \frac{5}{12}, \theta \in \left(\pi, \frac{3\pi}{2} \right)$
d. $\sin \theta, \tan \theta, \cot \theta$ and $\operatorname{cosec} \theta$ if $\cos \theta = -1, \theta \in \left(\frac{\pi}{2}, \frac{3\pi}{2} \right)$

E - ALGEBRAIC OPERATIONS WITH EXPONENTIALS, LOGARITHMS, ETC

1. Express the following as a power of 2.

$$\begin{array}{lll} \text{a.} & \frac{2}{\sqrt{2}} & \text{b.} \quad \sqrt[4]{8} \quad \text{c.} \quad 2.4^{2x} \\ \text{d.} & 2^{x+y} \div 2^{x-y} & \text{e.} \quad (2^x)^x \quad \text{f.} \quad \frac{2^5 \cdot 2^{1/5} \cdot 2^{3/5}}{2^{-1/5}} \end{array}$$

2. Simplify.

$$\begin{array}{llll} \text{a.} & 3a^4b^3 \cdot 2a^3b^{-2} & \text{b.} & \frac{12x^{2n+3}}{4x^{n-1}} \quad \text{c.} \quad (3x^2y^{-3}z^0)^{-4} \quad \text{d.} \quad (3^{x+1})^{x+1} \\ \text{e.} & (x^{1/y})^{1-y} & \text{f.} & \sqrt{9x^2} \quad \text{g.} \quad (4a^{-2}b^6)^{-\frac{1}{2}} \quad \text{h.} \quad \left(\frac{\sqrt{x}}{\sqrt{x-5}}\right)^{-\frac{1}{2}} \\ \text{i.} & \frac{7^{-1}x^{1/4}}{3^{-2}y^{-1}} \div \frac{2(3^{-1/2})^2}{(49y)^0} & \text{j.} & \frac{(2^{n-1})^{n-1}}{(2^{n+1})^{n+1}} \quad \text{k.} \quad \left(\frac{16a^3}{81a^{-1}}\right)^{-\frac{3}{4}} \\ \text{l.} & \frac{12^n \times 8^{n-1} \times 3^{n+1}}{24 \times 6^{n-2} \times 2^n} & \text{m.} & \sqrt[3]{a} \cdot \sqrt{a} \quad \text{n.} \quad \frac{(3^n)^{1-n}}{6^{-1} 3^{-n-1}} \div \frac{(3^{1-n})^{n+1}}{2^{-1} 9^{-n-1}} \\ \text{o.} & \frac{x^{8n-3}(xy^{1/2})^{-6n}x^3}{(xy)^{-4n}(x^{-1})^n} & \text{p.} & \frac{9^n - 3^{2n+1}}{(3^n)^2 - 3^{n+2} 3^n} \quad \text{q.} \quad \frac{5^x - 3^2 5^{x-2}}{5^{x+1} + 3.5^x} \\ \text{r.} & (a^{1/2} - b^{3/2})^2 & \text{s.} & \frac{a^{-1}}{x^{-1} - a^{-1}} \quad \text{t.} \quad \left(\frac{(3y)^{-2}}{3y^{-2}}\right)^{\frac{2}{3}} \end{array}$$

3. Express each logarithm as the sum or difference of simpler logarithms.

$$\begin{array}{llll} \text{a.} & \log \frac{xy}{z} & \text{b.} & \log \sqrt{\frac{x}{y}} \quad \text{c.} \quad \log(x^2y^3) \quad \text{d.} \quad \log \frac{1}{x^2y} \\ \text{e.} & \ln \frac{xy^2}{x+y} & \text{f.} & \ln \sqrt[3]{x^2+1} \quad \text{g.} \quad \ln \sqrt[3]{x^2y} \quad \text{h.} \quad \ln \sqrt{\frac{x^3}{y}} \\ \text{i.} & \log \frac{x}{yz} & \text{j.} & \ln \frac{(x+y)^3}{xy^2} \end{array}$$

4. Express each statement as a single logarithm with coefficient 1.

$$\begin{array}{lll} \text{a.} & 3 \ln x & \text{b.} \quad 3 \ln x - \frac{1}{2} \ln y \quad \text{c.} \quad \frac{1}{2}(\ln x - 3 \ln y) \\ \text{d.} & 2 \ln x + \ln(x+y) & \text{e.} \quad 2 \ln x - 3 \ln y + \ln z \quad \text{f.} \quad \ln(y-2) + \ln y - 2 \ln x \\ \text{g.} & \frac{1}{2}(\ln x - 5(\ln y - 3 \ln z)) & \text{h.} \quad \frac{1}{2}(\log x - 5 \log y - 3 \log z) \\ \text{i.} & \log 5 + 3 \log x - 2 \log 4 - \log y & \text{j.} \quad 3 \log(xy) - 2 \log x - \log y \end{array}$$

5. If $\log_a 2 = x$ and $\log_a 3 = y$, express each of the following in terms of x and y .

$$\text{a.} \log_a 27 \quad \text{b.} \log_a \frac{2}{9} \quad \text{c.} \log_a \sqrt[3]{6} \quad \text{d.} \log_a \sqrt{12a}$$

6. If $f(x) = 2^x$ determine the equations of g , h and k if

- (a) g is symmetrical to f about the x -axis.
- (b) h is symmetrical to f about the y -axis.
- (c) k is symmetrical to f about the line $y = x$.

7. If $\log_a b = 3$, find $\log_b a$.

8. Show that if $y = \frac{r}{1 + ce^{-at}}$ then $t = \frac{1}{a} \ln \frac{cy}{r-y}$.

9. If $f(x) = \log_b x$, find $f(\frac{1}{b})$ and $f(\sqrt{b})$.
10. If $\log_b \frac{1}{10} = -\frac{1}{2}$, find b .
11. Simplify
- | | | |
|--|--|---------------------------------------|
| a. $4(e^x)^4 + (x^e)^4$ | b. $\frac{1}{2}e^{2x} \cdot 4e^{3/x}$ | c. $\frac{14e^{3x}}{7e^x}$ |
| d. $\sqrt{e^x} \cdot \sqrt[3]{e^{-3x}}$ | e. $\ln(e^{3x})$ | f. $e^{\ln 7x}$ |
| g. $\ln(9e^x \cdot 10e^{2x})$ | h. $e^{\ln(\frac{x^2}{y})}$ | i. $3 \ln x + \ln \frac{1}{x}$ |
| j. $\ln e^2 + e^{-\ln x}$ | | |
12. Solve for x without using a calculator.
- | | | |
|-----------------------------|---------------------------------------|---|
| a. $\ln e^{4x} = 12$ | b. $e^{4x} = 12$ | c. $\ln 4x = 12$ |
| d. $e^{2 \ln 3} = x$ | e. $\ln x = 3 \ln 2 - 2 \ln 3$ | f. $\frac{1}{2} \ln x = 1 - \ln 2$ |

F - GRAPHS

1. Sketch the graphs of the following functions.

a. $f(x) = 2x + 3, \quad x \in [-4, 3]$	b. $f(x) = 2x - 1 $
c. $f(x) = x + 1$	d. $f(x) = \begin{cases} x^2, & x \leq 1 \\ -x + 2, & x > 1 \end{cases}$
e. $f(x) = x^2 - x - 6$	f. $f(x) = \sqrt{9 - x^2} \quad x \in [-3, 3]$
g. $f(x) = 2 \sin x + 1 \quad x \in [-2\pi, \pi]$	h. $f(x) = \cos \frac{x}{2} \quad x \in [0, 2\pi]$
i. $f(x) = \tan 2x \quad x \in \left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$	

2. Determine the domain of the function and sketch the function.

(After you have done Units 1.6 to 1.9)

a. $f(x) = \ln x$	b. $g(x) = \ln(-x)$	c. $h(x) = \ln x $
d. $l(x) = \ln x $	e. $m(x) = \ln(1 + x)$	f. $m(x) = 1 + \ln x$

3. Determine the domain of the function and sketch the function.

(After you have done Units 1.6 to 1.9)

a. $f(x) = e^x$	b. $g(x) = e^{-x}$	c. $h(x) = e^{ x }$
d. $l(x) = e^x $	e. $m(x) = e^{1+x}$	f. $n(x) = 1 + e^x$

G - GRAPHS (After you have done Units 1.7, 1.9, 1.10 and 3.7)

Sketch the graphs of the following pairs of functions on the same set of axes.

1. $f(x) = \sinh x$ and $g(x) = 2 \sinh x + 1$

2. $f(x) = \cosh x$ and $g(x) = \cosh \frac{x}{2}$
3. $f(x) = \tanh x$ and $g(x) = \tanh 2x$
4. $f(x) = \ln x$ and $g(x) = \ln 2x$
5. $f(x) = \ln x$ and $g(x) = \ln(x + 2)$
6. $f(x) = \ln x$ and $g(x) = 2 \ln x$
7. $f(x) = e^x$ and $g(x) = e^{2x}$
8. $f(x) = e^x$ and $g(x) = \ln x$
9. $f(x) = \sin x$ and $g(x) = \arcsin x$
10. $f(x) = \cos x$ and $g(x) = \arccos x$
11. $f(x) = \tan x$ and $g(x) = \arctan x$
12. $f(x) = \arcsin x$ and $g(x) = \arcsin(x + 1)$.

H - LIMITS

Determine the following limits (if they exist).

- | | | |
|--|--|--|
| 1. $\lim_{x \rightarrow 0} \left(x + \frac{1}{x}\right)$ | 2. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1}$ | 3. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ |
| 4. $\lim_{t \rightarrow 0} \frac{t^2 - t}{t}$ | 5. $\lim_{x \rightarrow 2} \frac{\pi}{2}$ | 6. $\lim_{x \rightarrow 0} x $ |
| 7. $\lim_{x \rightarrow e} \ln x^2$ | 8. $\lim_{x \rightarrow \frac{\pi}{2}} \tan x$ | 9. $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x)$ |
| 10. $\lim_{u \rightarrow 2} (u^3 + 4u + 1)$ | 11. $\lim_{x \rightarrow \infty} \left(x + \frac{1}{x}\right)$ | 12. $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^3 + 2x^2 + 1}$ |
| 13. $\lim_{x \rightarrow 0^+} \ln x$ | 14. $\lim_{x \rightarrow 3^-} \frac{-1}{\sqrt{3 - y}}$ | 15. $\lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 2x + 1}$ |

I - DIFFERENTIATION

Differentiate the following functions.

1. $f(x) = 2x^4 - 3x^2 + 5x + 2$
2. $f(x) = x^{-1/2} + x^{1/2}$
3. $f(x) = x^2 - 3x^{7/3}$
4. $f(t) = t + 1 + \frac{1}{t^2}$
5. $f(x) = \frac{12}{x} - \frac{4}{x^3} + \frac{1}{x^4}$
6. $f(x) = \sqrt[3]{x} + \frac{1}{x^3}$
7. $f(x) = 3x^2 + \frac{1}{3}x^{-2} + x$
8. $f(x) = \sqrt{x} - \frac{1}{\sqrt{x}}$
9. $f(x) = x^4 - 8x^3 + 2x^2 - x + 1$
10. $f(y) = 7y^3 - 5y^2 + 3y - 17$
11. $f(x) = 9x^{-3} + 2x^{-2} - 14$
12. $f(x) = -2x^4 + x^{-2} - 3x^{3/4}$
13. $f(x) = 4x^{-2} - 7\sqrt{x} + 8x^3 + 5$
14. $f(x) = 3x^3 + 2x^2 - x + 1$
15. $f(x) = x^3 - 3x - 2x^{-4}$
16. $f(x) = x + 3 + \frac{4}{x}$
17. $f(t) = 12t^4 + 3t^3 + 5t^{-2} - 4$
18. $f(x) = 4x^3 - 7x^2 + 8x - 6$
19. $f(u) = -3u^{-3} + 4u^2 + \frac{1}{u^2}$
20. $f(x) = x\sqrt{x} + \frac{1}{x^2\sqrt{x}}$
21. $f(x) = (x^3 + 7x^2 + 5)^5$
22. $f(x) = \sqrt{x^4 + x^2 + 2}$
23. $f(t) = \sqrt{t^2 + 2t + 3}$
24. $f(x) = x^2(x^3 - 1)$
25. $f(x) = x\sqrt{1 - x^2}$
26. $f(x) = (2x - 4)(3x^2 + 2)$
27. $f(x) = \sqrt{x + 1} (x^2 + 1)$
28. $f(x) = (x^3 - 3x)^4$
29. $f(x) = (2x - 3)^3(4x + 2)^2$
30. $f(x) = (7x + 3)^2(3x^2 - 14x + 5)$
31. $f(y) = (8y^3 - 2)(3y^2 - 5y + 10)^2$
32. $f(x) = (2x^3 - 3)^{2/3}$
33. $f(x) = (2x - 3)(3x + 4)$
34. $f(x) = (2x^3 - 1)(x^4 + x)$
35. $f(x) = (2x + 1)^2(x^2 + 2)^3$
36. $f(x) = \sqrt{3x^2 + 1}$
37. $f(x) = (3x^2 - 2x + 1)^{1/2}$
38. $f(x) = \left(\frac{1}{x} + 3\right)(x^2 - 5)$
39. $f(t) = (2t - 5)^3$
40. $f(x) = (6x - 5)^{-3}$

- 41.** $f(x) = \frac{x^2 + x + 1}{x^2 + 1}$
42. $f(x) = \frac{2x + 3}{3x + 2}$
43. $f(u) = \frac{2u + 1}{3u - 5}$
- 44.** $f(x) = \frac{x^2 + 5x - 1}{x^2}$
45. $f(x) = \frac{(x + 1)(x + 2)}{(x - 1)(x - 2)}$
46. $f(x) = \sqrt{\frac{x^2 + 1}{x^2 + 4}}$
- 47.** $f(t) = \frac{t + t^3}{\sqrt{t}}$
48. $f(x) = \left(\frac{x + 1}{x - 1}\right)^2$
49. $f(x) = \frac{10}{\sqrt{x} + 4}$
- 50.** $f(x) = \frac{8}{4 + x^2}$
- 51.** $f(y) = \frac{4y}{y^2 + 1}$
52. $f(x) = \frac{3}{\sqrt{2x + 1}}$
53. $f(x) = \frac{3x^2 - 2x + 3}{4x^2 - 5}$
- 54.** $f(x) = \frac{(2x - 4)(3x + 5)}{2x^2 + 7}$
55. $f(x) = \frac{2x - 3}{x^2 + 2x}$
56. $f(t) = \frac{4 - 2t + 3t^2}{t^2 + 2}$
- 57.** $f(x) = 2x^3 + \frac{2 - x}{x^3}$
58. $f(x) = \frac{x - 1}{x + 1}$
59. $f(x) = \frac{x^2}{x^2 + 1}$
- 60.** $f(t) = \frac{1}{t^4 - 2t + 1}$
- 61.** $f(x) = \sin(x^2 + 2)$
62. $f(x) = \sin(3x)$
63. $f(y) = \cos(3y)$
- 64.** $f(x) = \cos(x^4 + 7)$
65. $f(x) = \tan(\sin x)$
66. $f(t) = \tan(t^2 + 5)$
- 67.** $f(x) = \sin(3x + 2)$
68. $f(x) = 2 \sin x - \tan x$
69. $f(x) = 1 + x - \cos x$
- 70.** $f(x) = \frac{\sin x}{x}$
- 71.** $f(x) = \tan(2x - x^3)$
72. $f(\theta) = \sin(3\theta^2 - 2\theta + 1)$
- 73.** $f(x) = 3 \sin x - 2 \cos x$
74. $f(x) = -\cos(\pi x - 1)$
- 75.** $f(x) = x^2 \sin x$
76. $f(x) = \cos(\tan x)$
- 77.** $f(x) = 3 \sin x^2 + 2 \sin x - \sin 3$
78. $f(x) = \cos(3x) - \tan(3x)$
- 79.** $f(x) = \sin(2x + 3)$
80. $f(t) = \sin(2\pi t)$

81. $f(x) = \sin \frac{1}{\sqrt{x}}$ 82. $f(x) = \sin \frac{1}{x}$ 83. $f(x) = \cos(-x)$
84. $f(x) = \tan\left(\pi\left(\frac{1}{2} - x\right)\right)$ 85. $f(x) = \tan(x^2 + 1)$ 86. $f(t) = \tan(5t)$
87. $f(x) = x \cos x$ 88. $f(x) = \frac{\sin x}{1 - x}$ 89. $f(x) = 4 \sin^2(3x)$
90. $f(x) = \frac{\cos x}{\sin x}$ 91. $f(x) = (x^2 + 3) \sin x$ 92. $f(x) = \sqrt{\sin x}$
93. $f(x) = \cos^2 x^3$ 94. $f(x) = x \cos(5x)$ 95. $f(y) = \sin(2y) \cos(3y)$
96. $f(x) = \sin^2(3x) + \cos^2(5x)$
97. $f(x) = x^2 \tan \frac{1}{x}$ 98. $f(x) = (\sin x - x \cos x)^{-1}$ 99. $f(x) = \frac{\sin x + \cos x}{\tan x}$
100. $f(x) = \sin^2\left(x + \frac{1}{x}\right) + \cos^2\left(x + \frac{1}{x}\right)$
101. $f(x) = \frac{x^2}{\tan x}$ 102. $f(x) = \sqrt{\cos(2x)}$ 103. $f(x) = \sin^2(x - 3)$
104. $f(t) = \cos^3(t^2 - t)$ 105. $f(x) = \sin x \cos^2 x$ 106. $f(x) = \sin^{1/3}(2x)$
107. $f(x) = 4 \sin^7(2 - 4x)$ 108. $f(x) = 2 \sin x + 3 \sin^2 x$
109. $f(u) = 3 \cos^3 u + 4 \cos^2 u - 6$
110. $f(x) = 5 \cos^2(x + 2) + 3 \cos(x + 2) - 5$
111. $f(x) = 5x^2 - 3 \tan^2 x + \sin x$ 112. $f(x) = \frac{\tan x}{1 - \tan x}$ 113. $f(x) = (\sin(x + 1))^{3/2}$
114. $f(t) = \sin^2 t + \cos^2 t$ 115. $f(x) = \sin \sqrt{x} + \sqrt{\sin x}$
116. $f(x) = (\sin x)(\sin x + \cos x)$
117. $f(x) = 2 \sin x \cos x$ 118. $f(u) = \frac{1}{\sin u}$ 119. $f(x) = \cos^2 x \sin x$
120. $f(x) = \sqrt{1 - \sin^2 x}$
121. $f(x) = x + \tan^2 x$ 122. $f(x) = \frac{x}{\cos x}$ 123. $f(x) = \tan x \cos x$
124. $f(t) = \sin t \cos t$ 125. $f(x) = e^{x^2}$ 126. $f(x) = e^{5x}$
127. $f(x) = (x^2 + 3x)e^x$ 128. $f(x) = xe^x - e^{-x}$ 129. $f(x) = e^{x^2} \cdot e^{x+1}$
130. $f(x) = \frac{e^{x^2}}{e^{x-1}}$
131. $f(u) = e^{-u}$ 132. $f(x) = e^{\frac{2x}{3}}$ 133. $f(x) = e^{\sqrt{x}}$
134. $f(x) = e^{3x} + 2e^{2x} - 3e^x + 7$ 135. $f(x) = e^{x^2-2}$ 136. $f(x) = \frac{1 + e^{2x}}{2 - e^{2x}}$
137. $f(x) = e^{3x-1} - 4e^{-x}$ 138. $f(t) = \cos(e^t)$ 139. $f(x) = 3e^{2x} - 4e^x + 1$
140. $f(x) = e^{3 \cos(2x)}$
141. $f(x) = e^{-2x} + 4e^{-3x} + 7$ 142. $f(x) = e^{2x+1}$ 143. $f(x) = \frac{1}{2}e^{2x}$
144. $f(x) = e^{\sin x}$ 145. $f(x) = e^{2x}$ 146. $f(t) = 2te^t$
147. $f(x) = \frac{1}{1 - e^{-x}}$ 148. $f(x) = \frac{e^{-x}}{x}$ 149. $f(x) = x^2 e^{-x}$

150. $f(x) = e^{-\frac{1}{x^2}}$
151. $f(x) = e^{\sqrt{x^2+1}}$
152. $f(x) = \frac{e^x - e^{-x}}{2}$
153. $f(x) = e^{2x} \cos(3x)$
154. $f(x) = e^{\cos(4x)}$
155. $f(x) = x^2 \cdot 2^x$
156. $f(x) = 3^{5x}$
157. $f(x) = x^4 + 4^x$
158. $f(x) = 9^{-x}$
159. $f(x) = \tan 5^x$
160. $f(x) = 3^{4x+1} + 2^{4x+2}$
161. $f(y) = 3^{y^2+1}$
162. $f(x) = 2^x$
163. $f(x) = 2^{-x}$
164. $f(t) = (\frac{1}{2})^t$
165. $f(x) = e^x \ln x$
166. $f(x) = \ln(\sin x)$
167. $f(x) = \frac{1}{\ln x}$
168. $f(x) = \ln(3xe^{-x})$
169. $f(x) = \ln \frac{x-1}{x^2+1}$
170. $f(x) = \ln \frac{e^x}{1+e^x}$
171. $f(x) = \ln(e^{\sin 2x})$
172. $f(x) = \ln \sqrt{\frac{x}{x^2+1}}$
173. $f(u) = \ln(u^2)$
174. $f(x) = \ln \frac{10}{x}$
175. $f(x) = \ln 10^x$
176. $f(x) = \ln(3x) + 4 \ln x$
177. $f(x) = x^2 \ln(2x)$
178. $f(x) = \ln(x^{-1})$
179. $f(x) = x \ln x$
180. $f(x) = \ln \frac{1}{x}$
181. $f(y) = (\ln y)^3$
182. $f(x) = x \ln \sqrt{x}$
183. $f(x) = \ln(7x)$
184. $f(x) = \sqrt{\ln x}$

J - DIFFERENTIATION

Differentiate the following functions.

1. $f(x) = \sinh(x^2 + 4x)$
2. $f(x) = \sinh(7x)$
3. $f(x) = \cosh(x^4 + 5x + 7)$
4. $f(t) = \tanh(\sinh 2t)$
5. $f(x) = \tanh(x^2 + 5)$
6. $f(x) = \frac{\sinh x}{x}$
7. $f(y) = \tanh(2y - y^3)$
8. $f(x) = 3 \sinh x^2 + 2 \cosh x - \tanh 3$
9. $f(x) = 3 \sinh(4x) - 2 \cosh(5x)$
10. $f(x) = \cosh(\tanh x)$
11. $f(x) = \cosh \frac{1}{\sqrt{x}}$
12. $f(x) = \tanh \frac{1}{x}$
13. $f(t) = \sinh(-t)$
14. $f(x) = \frac{\cosh x}{1+x}$
15. $f(x) = 4 \cosh^2(3x)$
16. $f(x) = \sinh^2 x^4$
17. $f(x) = \ln(e^{\cosh 2x})$
18. $f(\theta) = \arcsin(2\theta)$
19. $f(x) = \arccos(x^2)$
20. $f(x) = x(\arcsin x)$
22. $f(y) = \arcsin(2y - 3)$
23. $f(x) = 2x(\arctan x)$
21. $f(x) = \arcsin \frac{2}{x}$
24. $f(x) = \arctan(5x)$
25. $f(t) = \arcsin(\sin t)$
26. $f(x) = \arctan(3x - 4)$
27. $f(x) = \arccos \frac{x}{4}$

K - IMPLICIT DIFFERENTIATION

The following equations define y implicitly as a function of x . Determine $\frac{dy}{dx}$ in terms of x and y in each case.

1. $x^2 + y^2 = 100$
 2. $x^3 - y^3 = 6xy$
 3. $x^2y + 3xy^3 - x = 3$
 4. $x^3y^2 - 5x^2y + x = 1$
 5. $x^2 = \frac{x+y}{x-y}$
 6. $\sqrt{xy} + 1 = y$
 7. $(x^2 + 3y^2)^{35} = x$
 8. $\cos xy = y$
 9. $\sin(x^2y^2) = x$
 10. $\tan^3(xy^2 + y) = x$
 11. $\sqrt{3 + \tan xy} - 2 = 0$
-

L - ANTIDERIVATIVES

Find the most general form of the function f satisfying the following.

1. $f'(x) = x^8$
 2. $f'(x) = \frac{1}{x^6}$
 3. $f'(x) = x^{5/7}$
 4. $f'(x) = \frac{4}{\sqrt{x}}$
 5. $f'(x) = \frac{1}{2x^3}$
 6. $f'(x) = x^3\sqrt{x}$
 7. $f'(t) = \sqrt[3]{t^2}$
 8. $f'(x) = x^3 - 2x + 7$
 9. $f'(x) = x^{-3} + \sqrt{x} - 3x^{1/4} + x^2$
 10. $f'(u) = u^{2/3} - 4u^{1/5} + 4$
 11. $f'(x) = \frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}$
 12. $f'(x) = x(1 + x^3)$
 13. $f'(t) = (1 + t^2)(2 - t)$
 14. $f'(x) = x^{1/3}(2 - x)^2$
 15. $f'(x) = \frac{1}{x^2} - \cos x$
 16. $f'(\theta) = 4\sin \theta + 2\cos \theta$
 17. $f'(y) = e^y$
 18. $f'(x) = xe^{x^2}$
 19. $f'(x) = \frac{2}{x}$
-

M - DIFFERENTIAL EQUATIONS

Find the solution of each of the following differential equations with initial values. (This means you have to find the function y .)

1. $\frac{dy}{dx} = \frac{x}{2}, y\left(\frac{1}{2}\right) = -1$
2. $\frac{dy}{dx} = -\frac{3}{2}x^2, y(-1) = -\frac{1}{2}$
3. $\frac{dy}{dt} = \sin t, y\left(\frac{\pi}{2}\right) = 3$
4. $\frac{dy}{dx} = e^x, y(0) = 4$
5. $\frac{dy}{dx} = \frac{1}{x}, y(1) = 2$
6. $\frac{d^2y}{dx^2} = 0, y'(2) = 1, y(2) = 2$
7. $\frac{d^2y}{dx^2} = \cos x, y'(0) = 1, y(0) = 2$
8. $\frac{d^2y}{dt^2} = e^t, y'(1) = e, y(1) = -4e$

N - INTEGRATION BY INSPECTION

Find the following integrals:

1. $\int e^{5x} dx$

2. $\int x^8 dx$

3. $\int \frac{1}{x^6} dx$

4. $\int t^{5/7} dt$

5. $\int \sqrt[3]{x^2} dx$

6. $\int \frac{4}{\sqrt{x}} dx$

7. $\int \frac{1}{2\theta^3} d\theta$

8. $\int x^3 \sqrt{x} dx$

9. $\int (x^3 - 2x + 7) dx$

10. $\int \left(x^{-3} + \sqrt{x} - 3x^{\frac{1}{4}} + x^2 \right) dx$

11. $\int \left(y^{\frac{2}{3}} - 4y^{\frac{1}{5}} + 4 \right) dy$

12. $\int \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x} \right) dx$

13. $\int x(1 + x^3) dx$

14. $\int (1 + u^2)(2 - u) du$

15. $\int x^{1/3}(2 - x)^2 dx$

16. $\int \left(\frac{1}{t^2} - \cos t \right) dt$

17. $\int (3 \sin x - 4 \cos x) dx$

18. $\int e^{-x} dx$

19. $\int 2xe^{x^2} dx$

20. $\int \frac{1}{2x} dx$

21. $\int e^{3t} dt$

22. $\int \frac{3}{x} dx$

23. $\int \frac{1}{2x + 1} dx$

24. $\int \sinh 4x dx$

25. $\int \sinh(4x + 6) dx$

26. $\int \cosh(2x + 3) dx$

27. $\int \tanh 3x dx$

28. $\int \frac{u^2}{u^3 - 4} du$

29. $\int \frac{5u^4}{u^5 + 1} du$

30. $\int \frac{t + 1}{t} dt$

31. $\int \frac{x^3}{x^2 + 1} dx$

32. $\int x^2(2 - x^3)^4 dx$

33. $\int xe^{2x^2} dx$

34. $\int \sin \theta (\cos \theta - 3)^3 d\theta$

35. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

36. $\int \frac{\sin \theta}{1 + \cos \theta} d\theta$

37. $\int \tan \theta d\theta$

38. $\int \frac{1}{x \ln x} dx$

39. $\int \frac{\ln y}{y} dy$

40. $\int (x + 1)\sqrt{x^2 + 2x} dx$